

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and operating large-scale distributed systems at **YouTube**, **Google**, and **Microsoft**. Managed teams of up to 15 engineers building video infrastructure, search ranking pipelines, and commerce platforms serving billions of users. Delivered systems processing **petabyte-scale daily data** with strict availability requirements. Seeking to apply expertise in backend systems and security to privileged access management development. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS, SQL

Distributed Systems: large-scale architecture, fault tolerance, load balancing, distributed databases, caching strategies

Backend Development: Python, Java, Go, C++, REST APIs, service-oriented architecture

Infrastructure: Kubernetes, cloud platforms, Terraform, monitoring systems, SRE practices

Data Engineering: data pipelines, streaming systems, analytics infrastructure, batch processing, AWS

Experience

Staff Software Engineer, YouTube

02/2025 - Present | Remote

- Leading video recommendation infrastructure development handling **billions of daily user interactions** with latency constraints under **100ms** using distributed caching and optimized data pipelines for real-time personalization.
- Migrated batch recommendation processing to streaming architecture, reducing system freshness from hours to minutes and improving user engagement metrics through more timely content suggestions.
- Established reliability practices for team of 8 engineers, achieving **99.99% uptime** for critical recommendation APIs through proactive monitoring, incident response, and post-mortem analysis.
- Designed multi-region load balancing strategy, reducing video serving latency by **40%** for global user base through intelligent traffic distribution, capacity planning, and geographic optimization.
- Mentored senior engineers on distributed systems design, promoting technical excellence and knowledge sharing across the organization through code reviews, architecture discussions, technical presentations, and documentation.

Staff Software Engineer/Manager, Google

03/2015 - 04/2024 | Remote

- Managed team of **12 engineers** developing search ranking infrastructure serving **billions of daily queries** across **190 countries**, coordinating across multiple time zones and product areas.
- Led global database migration to distributed relational system, eliminating regional latency issues and improving international search performance for users through optimized data placement.

- Reduced infrastructure costs by **\$2.4M annually** through capacity planning and resource optimization initiatives across distributed systems serving global traffic.
- Owned on-call rotation for critical search serving systems, maintaining **99.99% availability** during peak traffic periods and high-profile events through proactive monitoring.

Software Engineer, Google

02/2011 - 02/2015 | Remote

- Built data pipelines processing **terabyte-scale daily data** for advertising analytics using Python and C++, enabling real-time campaign performance insights for advertisers.
- Developed distributed caching layer reducing ad targeting latency by **60%** for real-time bidding systems through strategic data placement and cache invalidation strategies.
- Created monitoring infrastructure for pipeline health, cutting incident detection time from minutes to seconds through automated alerting and anomaly detection systems.
- Collaborated with machine learning teams to deploy ranking models at scale, integrating automated training pipelines with serving infrastructure for continuous improvement.

Software Design Engineer II, Microsoft

10/2007 - 12/2011 | Remote

- Designed authentication systems for enterprise identity platform supporting **hundreds of millions of users**, implementing secure credential management and access controls for global organizations.
- Built API layer for cloud identity management using C# and enterprise frameworks, enabling third-party application integrations through standard authentication protocols and OAuth implementations.
- Developed multi-tenant isolation mechanisms ensuring security compliance for enterprise customers with strict data governance and regulatory requirements across different regions.
- Optimized database operations reducing authentication latency by **35%** for high-traffic scenarios through query optimization, indexing strategies, and connection pooling improvements.

Software Design Engineer Intern, Microsoft

06/2006 - 08/2006 | Remote

- Built test automation framework for operating system components, reducing regression testing time by **25%** through automated test generation and execution pipelines.
- Developed performance analysis tools identifying bottlenecks in system-level operations, providing actionable insights for optimization and improved system efficiency.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007